

Insulin Resistance Diet

Seventy percent of individuals with insulin resistance (elevated insulin, HgA1c or fasting blood sugars) have atherosclerosis irregularities presenting on special heart imaging, with the overall risk of coronary heart disease death from metabolic syndrome 4.2 times greater. Combining imaging studies such as Carotid Imaging and Calcium CT scores with blood labs for cholesterol, HgA1c, fasting insulin and glucose, lipoprotein a, apolipoprotein B, oxidized LDL, homocysteine, and high sensitivity C-reactive protein increases the odds of predicting future events and risks. Because of your individual family history, or irregular imaging and blood studies, we are recommending the following lifestyle and diet adjustments focused on reducing future disease risks. This approach allows for a quick start to help address diabetes and prediabetes concerns.

1. Rule number one for preventing and reducing risk of heart disease and stroke risk is to control blood sugars. This is best accomplished by consuming **less than ____ grams total daily from complex grains**. This is approximately equivalent to 2 servings daily of grains such as bread, pasta, or cooked grains. One of the best ways to learn how your individual diet is affecting your blood sugars is to track your daily blood sugars using a continuous glucose monitor or CGM. These tools dramatically add to your awareness of how diet choices affect sugars and your risk for diabetes and thus heart disease.

If you are following a Carnivore, Paleo, Whole 30, Pegan, or Keto diet then this will be easy because complex carbohydrates from grains are a reduced part of the diet already. If you want to jumpstart your low complex carb diet, you can experiment with any of the above mentioned diets for any length of time desired.

Another obvious means to avoiding grains is fasting, particularly intermittent fasting. By eating in a smaller window of the day you will be likely skipping at least 1-2 meals on average. Of course when you don't eat anything you are not eating grains (makes it easy anyway).

2. Rule number two is to reduce processed foods by increasing your intake of nutrient rich foods in two ways: first increase intake of high fiber foods (found in fruits and vegetables) and second, increase nutrient dense essential macronutrients such as protein and fats. Carbohydrates as found in grains, which break down into simple sugars, are not an essential nutrient, meaning our bodies do not require them to maintain health or normal metabolic function. Thus they can be eliminated or at minimum greatly reduced. Focus on organic products free of pesticides. Visit www.ewg.org for lists of best organic choices.

When we increase the consumption of fruits and vegetables (and whole grains) we naturally increase the amount of fiber in the intestinal tract. Fiber acts as the most effective method to support a healthy microbiome and reduces multiple metabolic risks. Try to consume up to 40 grams of fiber daily. Adding additional fiber may be an option but should be done slowly to allow the body to adjust to the changes occurring in the gut microbiome.

Individual needs may vary and will be based on weight and physical activity, but try to consume at least 60-120 grams of protein daily in divided doses. Choose organic products free of antibiotics and hormones. Avoid pork for the most part. Limit exposure to environmental heavy metals particularly mercury as found in fish such as Tuna, Mahi mahi, Cod, Salmon and others. Visit www.ewg.org/tunacalculator/ for safe dosing guidelines.

Consume healthy essential oils, such as high polyphenol extra virgin olive oil (EVOO), avocado, nuts, and seeds. Coconut oil is full of healthy fatty acids to help induce ketosis, including keto-friendly medium-chain triglycerides (MCTs). Consider wild-caught seafood rich in healthy fats like salmon, sardines, and mackerel with grass-fed meat, in moderation

3. Rule number three, avoid the bad stuff including excessive Alcohol, soda, added sugars (<20 grams daily), and hydrogenated fats

Your daily carbohydrate limit is: _____
grams of carbohydrates – grams of fiber = total carbohydrates to count for the food

PROTEIN

- **Fish: aim to eat around 2-3 times per week**
 - Wild Alaskan salmon (canned too), herring, wild cod, trout, sardines, halibut, tilapia
 - Avoid these fish due to high mercury and toxin content
 - Tuna, farm-raised salmon (Atlantic salmon), catfish, mackerel, sturgeon, swordfish, shark
 - Other seafood OK to eat:
 - shrimp, clams, mussels, crab, lobster
- **Meat: try not to include red meat in more than 3 meals per week**
 - When eating meats they should be organic, pasture raised, grass-fed/grass-finished. Avoid cheaper agri-industry meat and highly processed meats with nitrates and other chemicals.
 - Chicken, turkey, beef, buffalo, venison, lamb, pork
 - Best to cook by roasting or baking
- **Plant based sources:**
 - Nuts and nut butters
 - Almonds, walnuts, peanuts, sunflower seeds, pumpkin seeds, hazelnuts, brazil nuts, pecans
 - Avoid cashews and nut butters with added sugar and partially hydrogenated oils
 - Beans and lentils (**watch portion sizes because there still can be a fair amount of carbs)
 - Use Beano if you have difficulty digesting, start with smaller portions to assess tolerance level and gradually increase. Aim for at least ½ cup per day.
 - Canned organic beans (Eden Organics uses BPA free cans)
 - Great source of fiber!!
 - Soy (should be organic non-GMO)
 - Edamame, tofu, tempeh
 - Avoid soy protein isolate, it's highly processed
 - Flax and chia seeds
- **Eggs**
 - Organic Omega-3 or Pasture-raised are the best
 - Cook by hard or soft boiling, sunny side up, or poached
- **Dairy: buy organic if possible and products made from grass-fed/grass-finished cows**
 - Cheese should be lower fat products
 - Yogurt- plain, you can add your own fruit. Avoid the ones with sugar added.
 - Avoid: creams due to high fat content, milk, cottage cheese

VEGETABLES

- Aim to eat 2-5 cups a day. A large salad for one meal per day is a great goal.
- Stir fry, steam, or bake- Avoid canned, boiled, or microwaving
- Emphasize dark leafy greens- romaine, spinach, kale, arugula, cabbage, collard greens, beet greens, mustard greens, parsley
- Broccoli, cauliflower, brussel sprouts, radish, cabbage (cruciferous)
- Asparagus, celery, cucumber, zucchini, peppers, garlic, chives, and many more!
- AVOID due to high carb content: beets, carrot, tomatoes, winter squash, eggplant, potatoes, yams, onions

GRAINS

- This is your biggest restriction, avoid as much as possible
- Almond meal is a low carb alternative
- Avoid: cereals, pasta/noodles, bread, pancakes, pastries, rolls, rice, corn
- When eating always use 100% whole grains: Brown rice, quinoa, oats, whole wheat, spelt, amaranth, buckwheat

OILS/FATS

- Organic unrefined coconut oil
- Organic extra virgin olive oil
- Walnut oil
- Flax oil (use raw- do not cook)
- Sesame oil
- Refined or high oleic/monounsaturated safflower oil
- Organic butter (best from grass-fed/grass-finished cows)
- Avocados and avocado oil
- Flax and chia seeds
- Avoid canola oil, margarines, anything with partially hydrogenated oils (trans fats, vegetable fat/shortening), fully hydrogenated, and interesterified oils
- Refrigerate all oils except, olive oil, avocado, and coconut oil

FRUITS

- ¼ cup to ½ cup max berries per day is the best option
- OR one portion of other fruits are ok, but no bananas (too high in sugar)
- Fruits in general should be limited due to high carb/sugar content

BEVERAGES

- **Water**- half your body weight in ounces every day!
- **Alternative milks**- (UNSWEETENED) almond, soy, coconut, rice, hazelnut, hemp, etc.
- **Fresh made veggie juice**- unlimited
- **Tea**- herbal, green, limit black tea
- **Coffee**- limit to a max of 1-2 cups per day
- **Alternative sodas**- Zevia, Virgil Zero, Blue Sky Free
- **Alcohol**- none or one drink 1-2 times a week max
- Avoid- sweetened beverages: soda, sweetened iced tea drinks, sweetened fruit drinks, Gatorade, and anything with artificial coloring

SWEETS/SUGAR

- Natural sweeteners/sugar alternatives are OK
 - Stevia (liquid or powdered form), Truvia, Erythritol, Xylitol
- AVOID: white/brown sugar, honey, agave, and artificial sweeteners
 - Sucrose, cane juice, glucose, fructose, high fructose corn syrup, dextrose, maltose, corn syrup, malted barley syrup, turbinado sugar, molasses, 100% maple syrup, date sugar, brown rice syrup, coconut sugar